

Physical Chemistry (I)

Lecture 12



In the Last Class

- Helmholtz and Gibbs energies
 - Definitions
 - Meanings
- Thermodynamics potentials
 - Differential forms
 - Properties of exact differentials
- Thermochemistry of Gibbs energy

Helmholtz and Gibbs Energies

$$\text{Helmholtz energy } A = U - TS$$

$$\text{Gibbs energy } G = H - TS$$

Criteria of spontaneity:

$$dA_{T,V} \leq 0$$

(At constant T and V , $dA \propto -dS_{\text{total}}$)

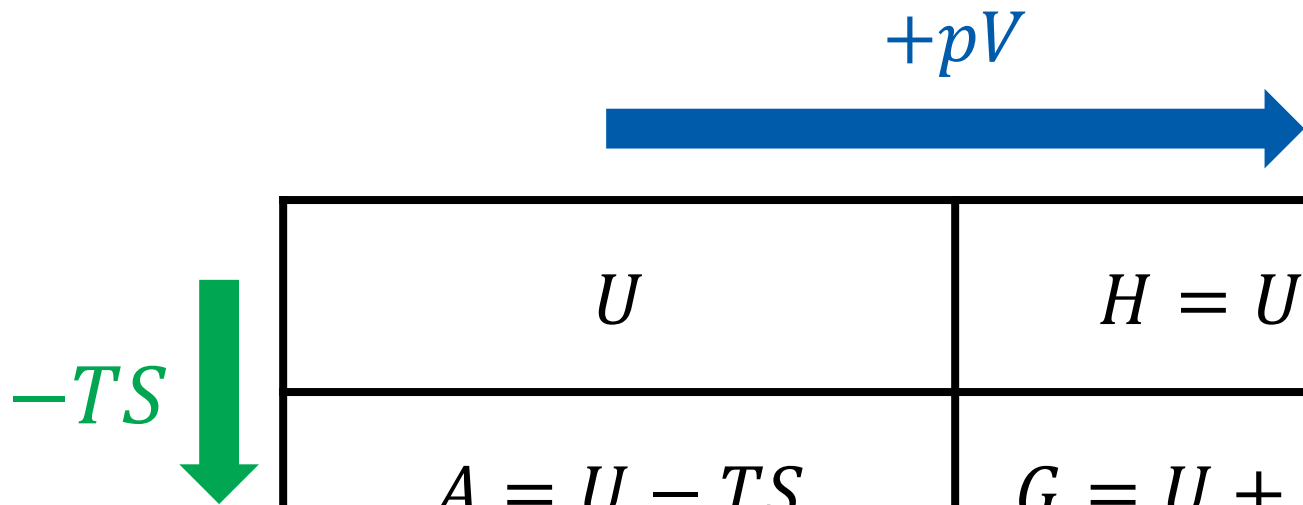
$$dG_{T,p} \leq 0$$

(At constant T and p , $dG \propto -dS_{\text{total}}$)

So, What Do They Mean?

- Helmholtz energy = maximum work
- Gibbs energy = maximum non-expansion work

Thermodynamic Potentials



U	$H = U + pV$
$A = U - TS$	$G = U + pV - TS$

Natural Independent Variables

- Internal energy $U = U(S, V)$

$$dU = TdS - pdV$$

- Enthalpy $H = U + pV = H(S, p)$

$$dH = TdS + Vdp$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$dA = -SdT - pdV$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$dG = -SdT + Vdp$$

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T} \right)_p = -\frac{H}{T^2}$$

At constant T ,

$$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_p = -\frac{\Delta H}{T^2}$$

Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $dU = TdS - pdV$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $dH = TdS + Vdp$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $dA = -SdT - pdV$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $dG = -SdT + Vdp$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

Thermodynamic Eq. of State

$$p = T \left(\frac{\partial p}{\partial T} \right)_V - \pi_T$$

Thermochemistry of G

$$G = H - TS$$

At constant T ,

$$\Delta G = \Delta H - T\Delta S$$

We can define the **standard Gibbs energy of reaction**:

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

and the **standard Gibbs energy of formation**:

$$\Delta_f G^\ominus = \Delta_f H^\ominus - T\Delta_f S^\ominus$$

Standard Reaction Potentials

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f H^\ominus(j)$$

$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

$$\Delta_r G^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f G^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f G^\ominus(j)$$

Pressure Dependence of G

$$\Delta G_m = \int_{p_i}^{p_f} V_m dp$$

1. Condensed phase: $V_m \approx \text{constant}$
2. Gas phase: $V_m = V/n = RT/p$

6. Phase Transitions

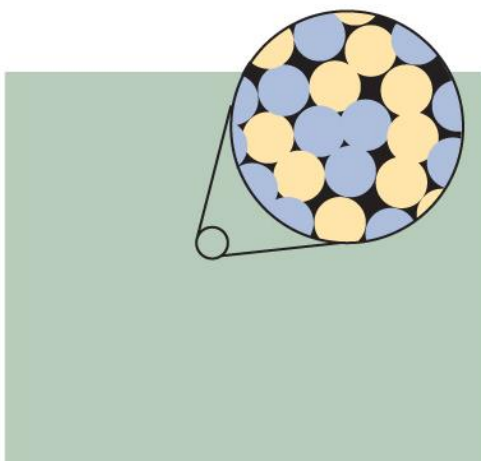
Chapter Overview

- Phase and chemical potential
 - Phase and phase transition
 - Chemical potential
 - Phase diagram and phase rule
- van der Waals fluid
 - Liquid-gas phase transition
 - Supercritical fluid
 - Critical point and the principle of corresponding states
- Thermodynamic aspects of phase transitions

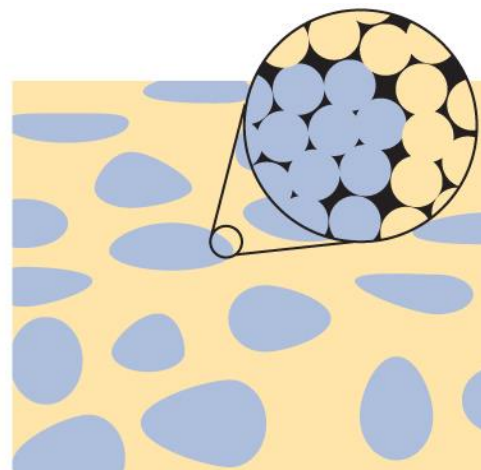
Phase

A form of matter that is **uniform** throughout
in **chemical composition** and **physical state**

Solution: Single Phase



solution
(one phase)



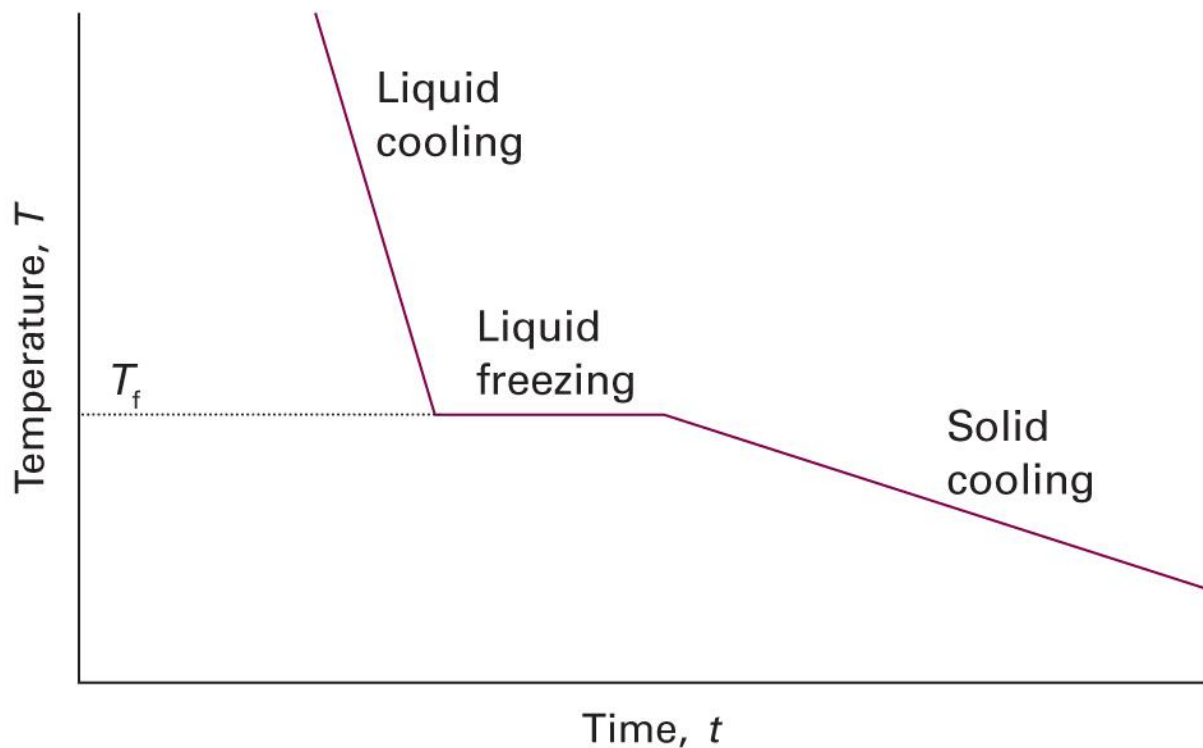
dispersion
(two phases)

Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Thermal Analysis



Two Phases at Equilibrium

$$dU = TdS - pdV$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: ?
 - Conjugate extensive variable: dn

Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn
- (Modified) differentials of thermodynamic potentials

$$dU = TdS - pdV + \mu dn$$

$$dH = TdS + Vdp + \mu dn$$

$$dA = -SdT - pdV + \mu dn$$

$$dG = -SdT + Vdp + \mu dn$$

3 Independent Variables

$$dU = TdS - pdV + \mu dn$$

This implies that $U = U(S, V, n)$, which leads to

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \quad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

Meaning of μ

$$dG = -SdT + Vdp + \mu dn$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

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Meaning of μ

$$g(T, p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n} \right)_{T, p}$$

However,

$$dG = -SdT + Vdp + \mu dn$$

gives

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T, p} = g(T, p) = \frac{G}{n}$$

(molar Gibbs energy)

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Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$

Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

Differentiating both sides,

$$dU = \mu dn + n d\mu + T dS + S dT - p dV - V dp$$

but we know that

$$dU = T dS - p dV + \mu dn$$

Thus,

$$S dT - V dp + n d\mu = 0 \text{ (Gibbs-Duhem equation)}$$

$$d\mu = -S_m dT + V_m dp$$

Two Systems in Contact

$$dG = -SdT + Vdp + \mu dn$$

At constant T and p ,

$$dG = \mu dn$$

For two systems in contact,

$$dG_1 = \mu_1 dn_1$$

$$dG_2 = \mu_2 dn_2$$

The total change in Gibbs energy is

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

Diffusive Equilibrium

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

If the two subsystems consist of the same type of particles and can exchange particles,

$$dn_1 + dn_2 = 0$$

Hence,

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

At equilibrium, $dG = 0$, and

$$\mu_1 = \mu_2$$

What If $\mu_1 \neq \mu_2$?

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

- If $\mu_1 > \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 < 0 \text{ and } dn_2 = -dn_1 > 0,$$

which means that the particles move from system 1 to 2

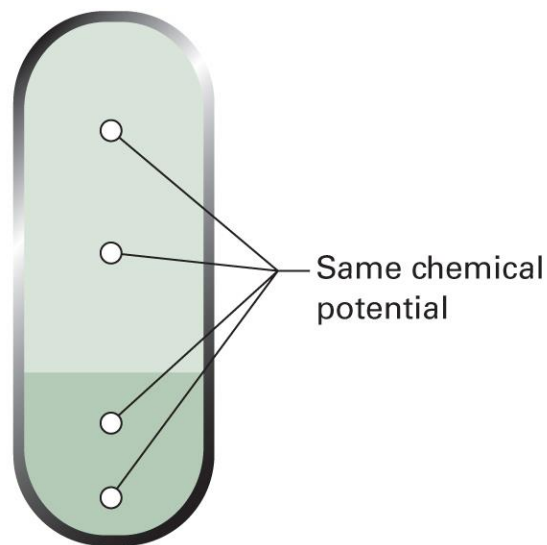
- If $\mu_1 < \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 > 0 \text{ and } dn_2 = -dn_1 < 0,$$

which means that the particles move from system 2 to 1

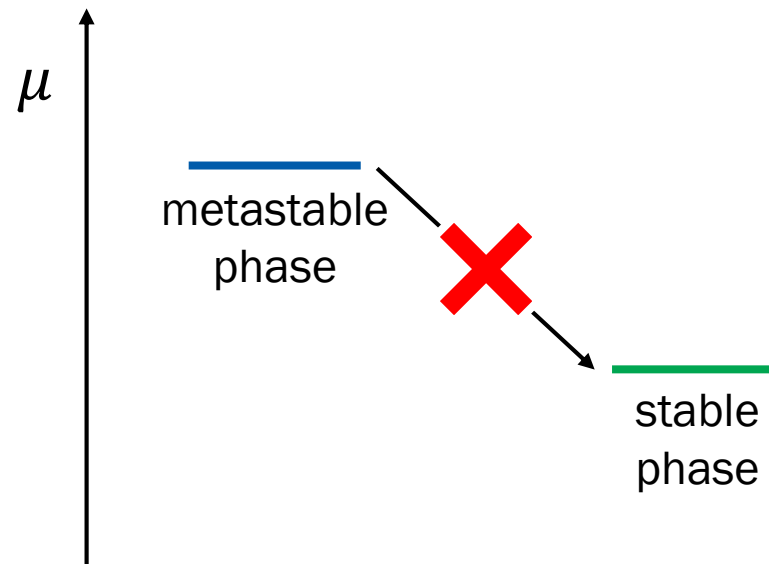
Criteria for Phase Stability

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system



Metastable Phase

Thermodynamically unstable phases that persist because the transition is kinetically hindered



Brief Illustration 4A.3

The standard molar Gibbs energy of formation of water vapour at 298 K (25 °C) is -229 kJ mol^{-1} , and that of liquid water at the same temperature is -237 kJ mol^{-1} . At this temperature, which one is spontaneous, condensation or vaporization?

At 298 K, $\mu_g = -229 \text{ kJ mol}^{-1}$ and $\mu_l = -237 \text{ kJ mol}^{-1}$

Since $\mu_g > \mu_l$,

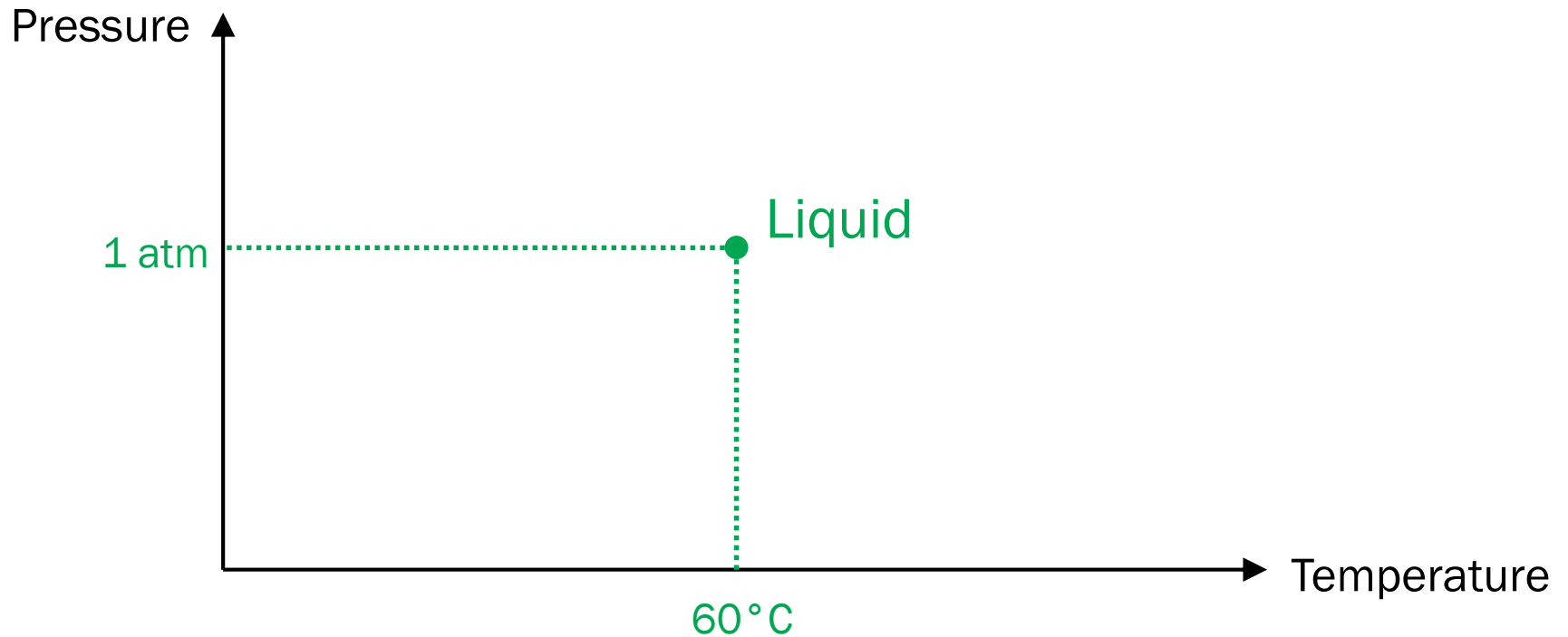
the transition from gas to liquid (condensation) is spontaneous.

Phase Diagram

A diagram that shows the regions at which its various phases are thermodynamically stable

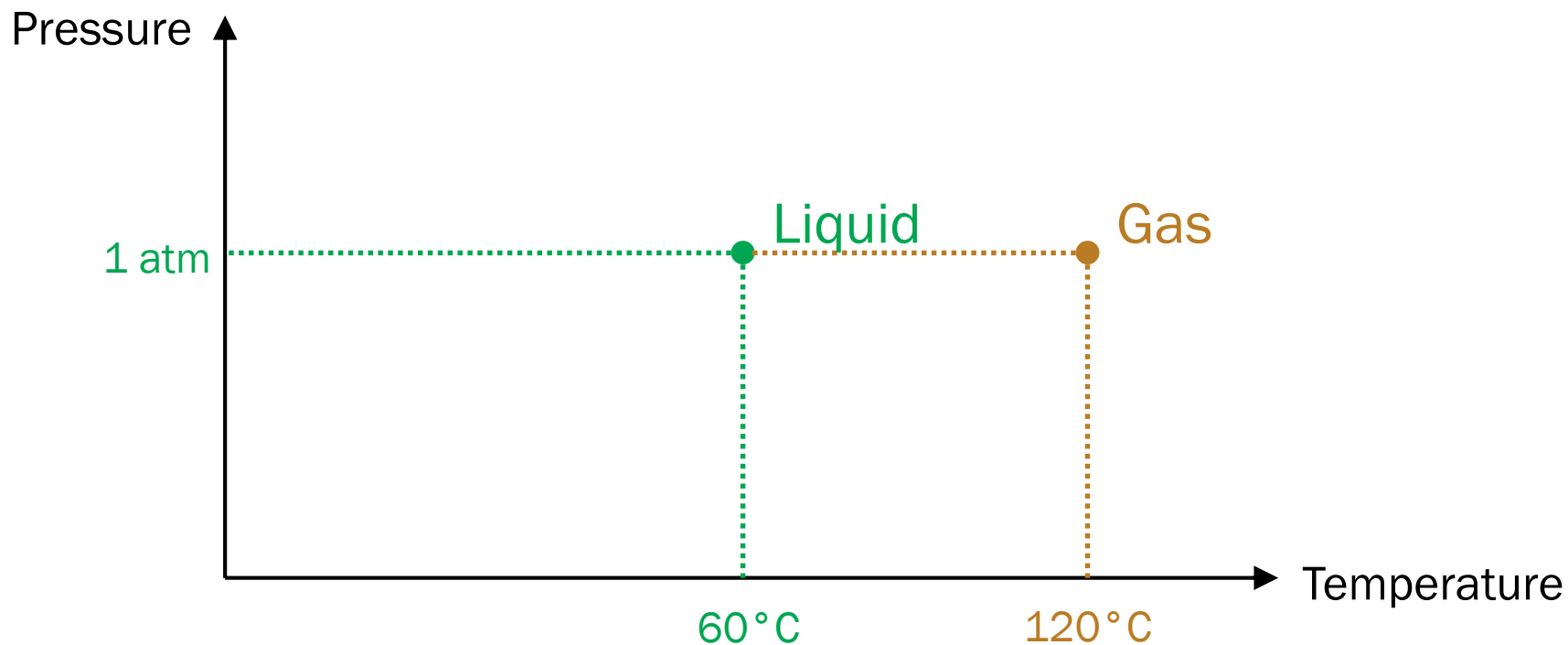
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Drawing Phase Diagram



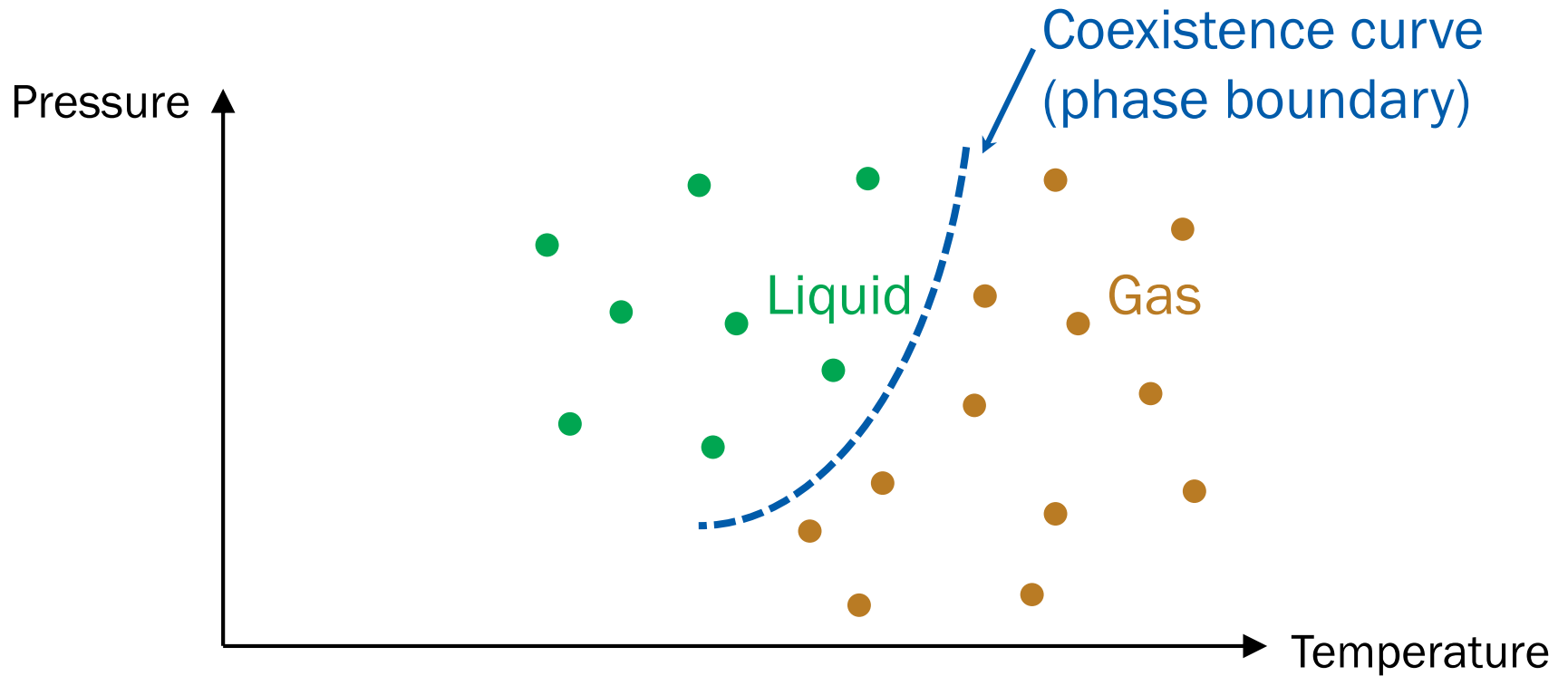
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Drawing Phase Diagram

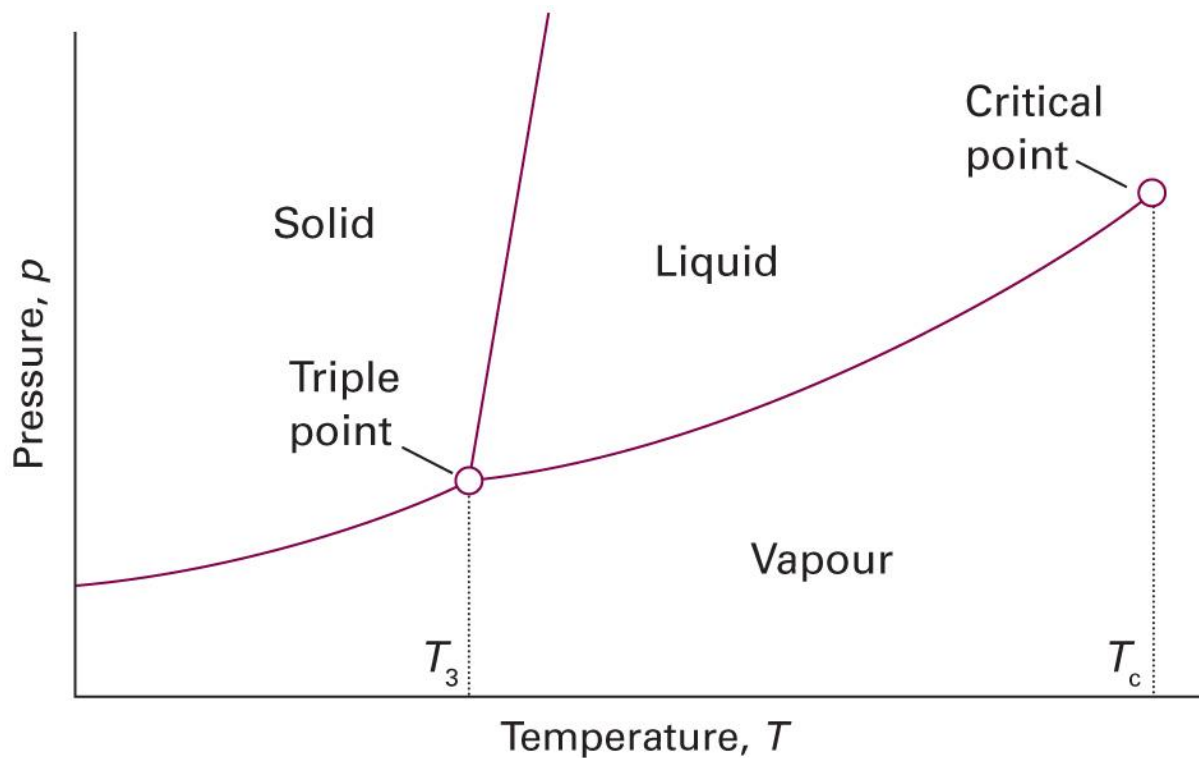


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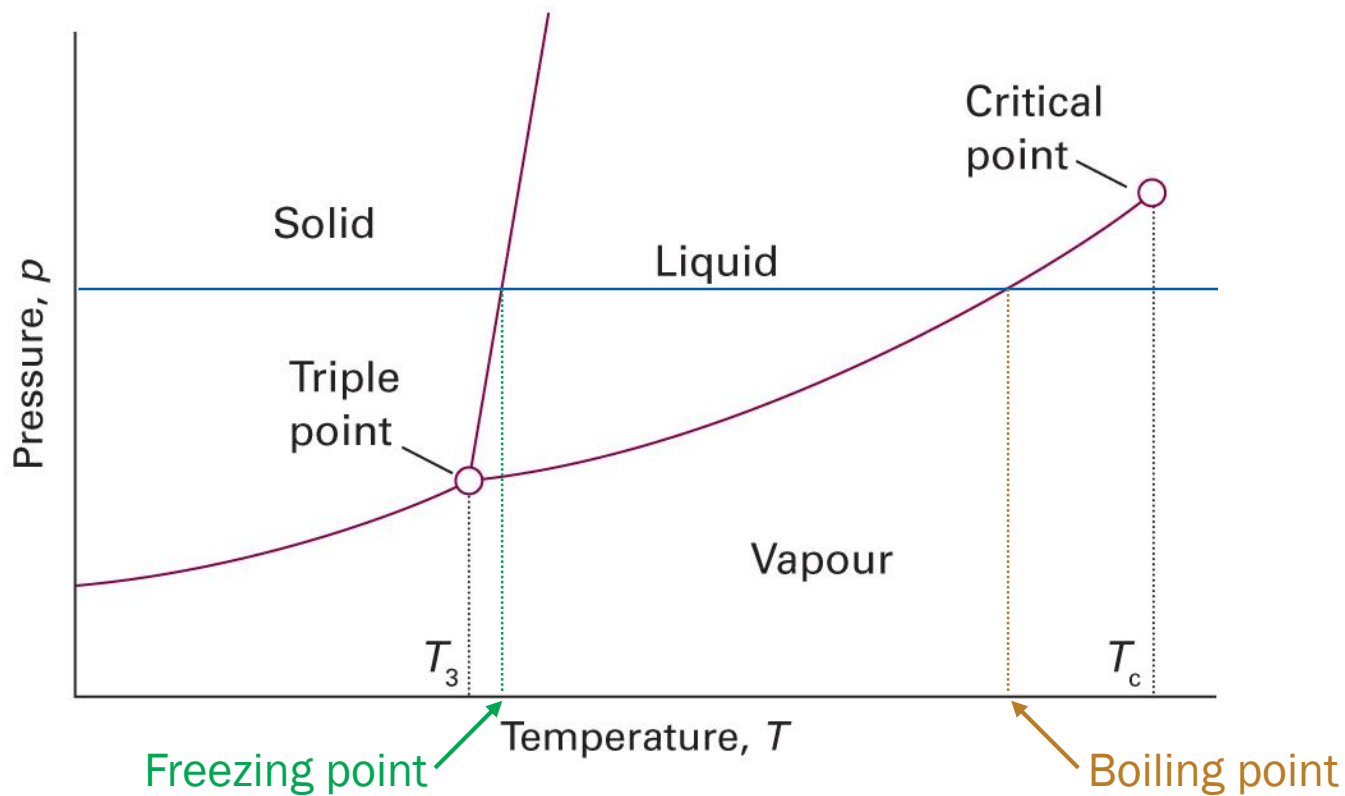
Drawing Phase Diagram



Typical Phase Diagram



Freezing and Boiling Points



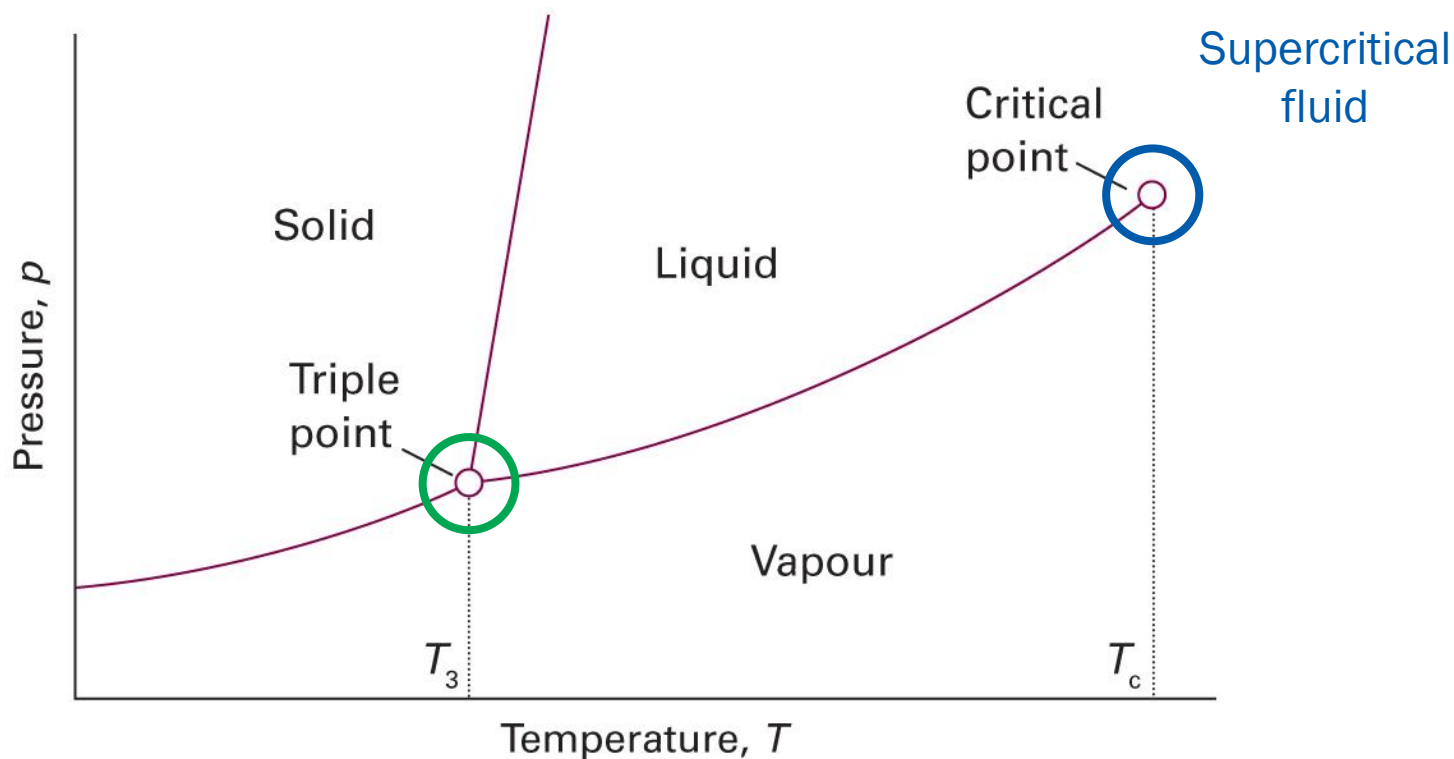
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.4)

Freezing and Boiling Points

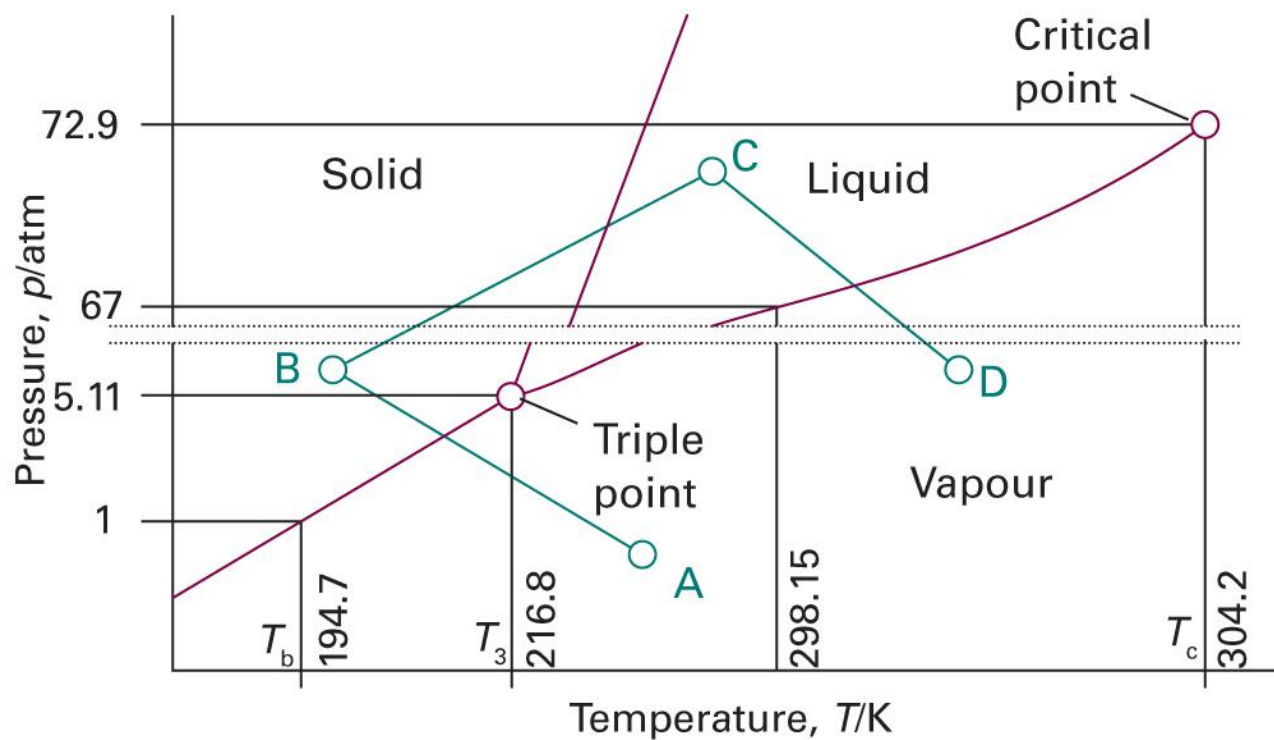
- Normal boiling point T_b : b.p. at 1 atm
- Standard boiling point: b.p. at 1 bar
- e.g. water: $T_b = 100^\circ\text{C}$, standard b.p. = 99.6°C

- Normal freezing point T_f : f.p. at 1 atm
- Standard freezing point: f.p. at 1 bar
- Usually the difference between the two is negligible

Two Characteristic Points

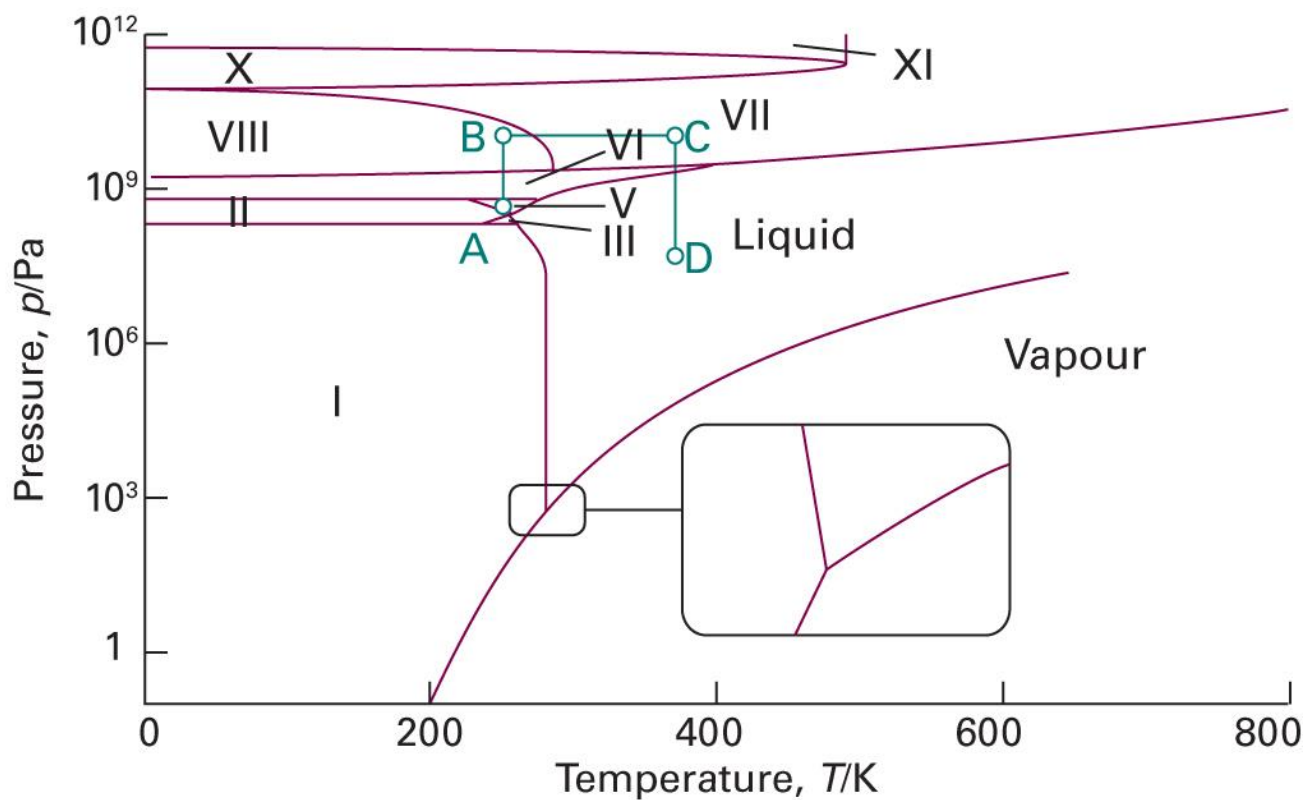


Carbon Dioxide



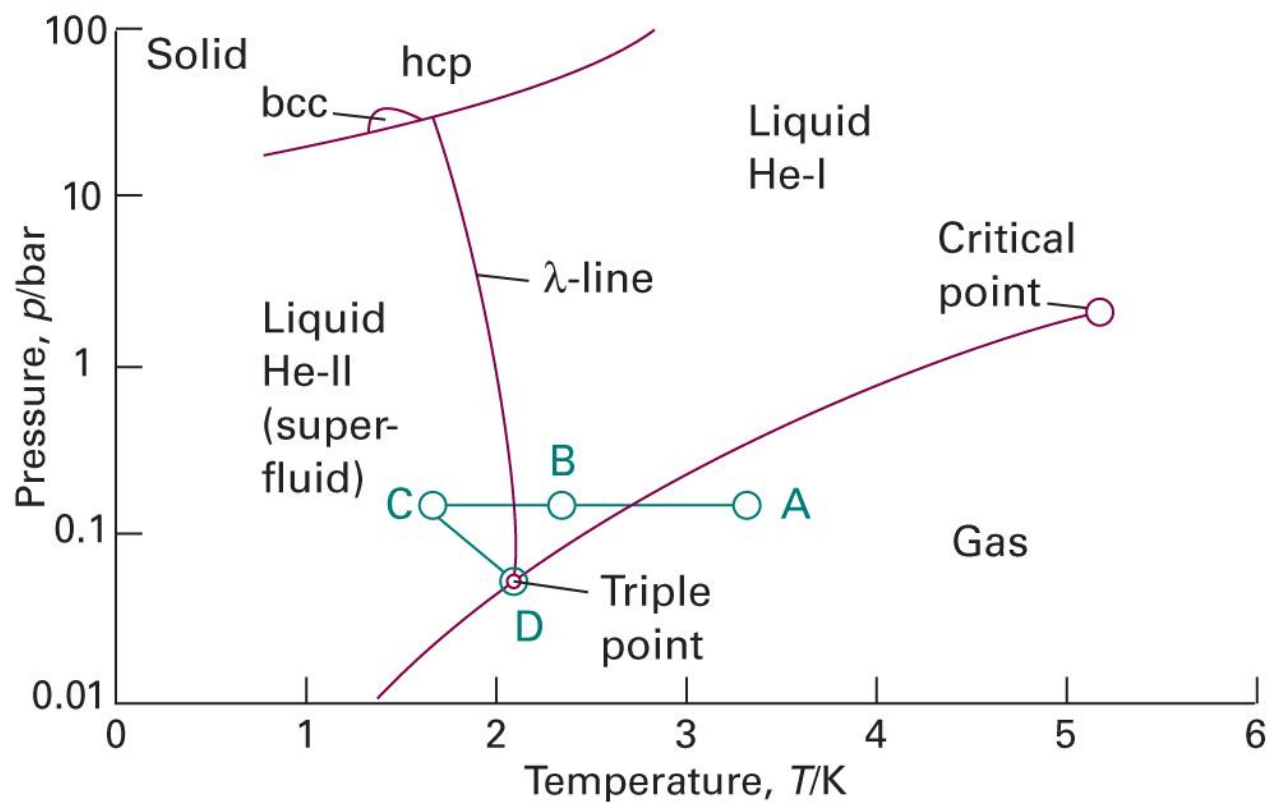
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.8)

Water



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.9)

Helium (^4He)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.11)

Phase Rule

“Can we have a quadruple point?”

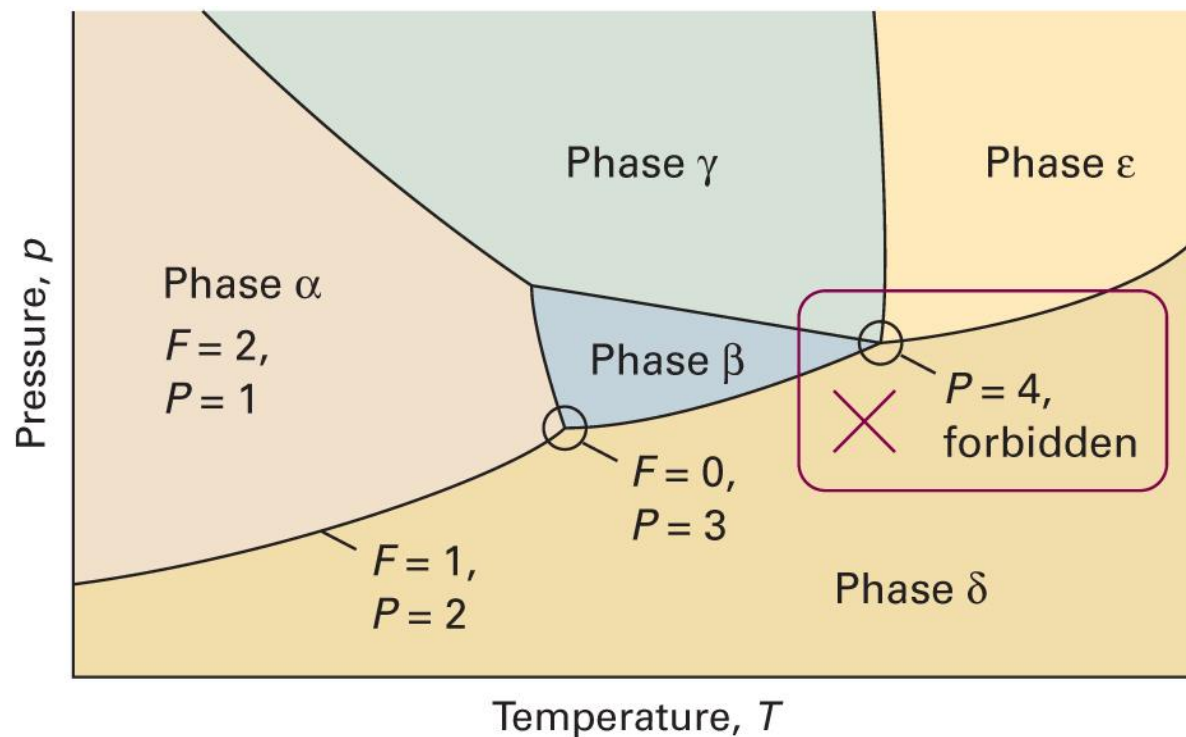
- **Phase rule:** a general relation between
 - Number of degrees of freedom F
 - Number of components C
 - Number of phases at equilibrium P
- Use the **criteria for phase stability** (same μ)
 - Recall that $\mu = g(T, p)$

Single Component ($C = 1$)

- $P = 1$: both T and p can be independently varied
 $\therefore F = 2$ (bivariant)
- $P = 2$: $\mu_1(T, p) = \mu_2(T, p)$
 $\therefore F = 1$ (univariant)
- $P = 3$: $\mu_1(T, p) = \mu_2(T, p)$ and $\mu_2(T, p) = \mu_3(T, p)$
 $\therefore F = 0$ (invariant)
- $P = 4$: two variables and three restrictions $\rightarrow$ impossible

In summary, $F = 3 - P$

Single Component ($C = 1$)



Multiple Components ($C > 1$)

- The total number of independent intensive variables
 - T and p : 2
 - $(C - 1)$ mole fractions (-1 due to normalization) for each phase: $P(C - 1)$
 - Total number = $P(C - 1) + 2$
- The total number of equations (restrictions)
 - $\mu_J(\text{phase 1}; p, T) = \mu_J(\text{phase 2}; p, T) = \dots$ for component J
 - Total number = $C(P - 1)$
- Therefore, $F = P(C - 1) + 2 - C(P - 1) = C - P + 2$